Thalamocortical Expansion and Representation Learning in the Visual System

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Abstract

The conversion of sensory inputs into robust cortical representations is a fundamental operation of the mammalian brain. In the early visual system, this process is characterized by a dramatic expansion in neuronal density as signals traverse from the Lateral Geniculate Nucleus (LGN) to the Primary Visual Cortex (V1). This "thalamocortical expansion" has long been hypothesized to facilitate the creation of overcomplete representations, yet the functional advantages of such high-dimensional coding remain difficult to quantify at scale. This report investigates the role of expansion in representation learning using a biologically constrained feedforward architecture. By simulating populations across a wide range of expansion factors, we demonstrate that a massive cortical over-representation is critical for the emergence of sparse, selective features that efficiently separate temporal variations in natural stimuli. Our findings provide support for the hypothesis that the thalamocortical expansion serves as a high-dimensional mapping stage necessary for complex metric learning in the visual hierarchy.

1 Introduction

The architectural layout of the early visual pathway provides a compelling example of biological specialized wiring. The process begins in the retina, where light is transduced into electrical signals by photoreceptors and subsequently processed by a mosaic of Retinal Ganglion Cells (RGCs). These cells form the bottleneck of the visual system, projecting through the optic nerve to the Lateral Geniculate Nucleus (LGN) of the thalamus. Biologically, the ratio of RGCs to LGN neurons is nearly 1:1, or characterized by a modest expansion. However, the subsequent projection from the LGN to the Primary Visual Cortex (V1) undergoes a radical transformation. In mammals, this thalamocortical pathway exhibits an expansion characterized by a power law of roughly $K = N^{3/2}$, where K is the number of V1 neurons and N is the number of LGN neurons. Clearly, a relatively small number of thalamic inputs are mapped onto a vast population of cortical neurons, reaching a 1:300 ratio in humans.

The functional necessity of this massive over-representation is a central theme in computational neuroscience. One prominent hypothesis is that this expansion allows the cortex to implement an overcomplete basis set for signal representation. Unlike a critically sampled system, an overcomplete system provides the flexibility to represent inputs with higher sparsity, a strategy computationally known to make downstream classification tasks significantly easier. By projecting the low-dimensional outputs of the LGN into a much higher-dimensional space in V1, the brain may be performing a non-linear mapping that simplifies the separation of overlapping stimulus features, effectively behaving like a kernel expansion in machine learning. This expansion is thought to be the foundation for the emergence of complex receptive field properties, such as orientation selectivity and direction sensitivity, which are absent in the circularly symmetric filters of the retina and LGN.

Furthermore, the geometry of this expansion is not random but follows precise topographic and probabilistic wiring rules. Neurons in V1 do not receive inputs from the entire LGN pool; instead, they integrate information from localized "neighborhoods," maintaining a retinotopic map while diversifying the feature

space. Understanding how this structural expansion relates to the quality of representation learning is the primary motivation for this project. We aim to quantify whether higher expansion factors directly lead to better category or temporal separation in a representation learning context, using naturalistic video stimuli that mimic the continuous stream of visual information encountered in biological environments.

2 Methodology

The trajectory of this project involved an iterative refinement of the model architecture to best capture biological principles while remaining computationally tractable for large-scale benchmarking.

Initially, effort was directed toward implementing the early visual system model proposed by Slapik & Shouval (2026). While the Slapik model offers insights into estimated representations created by each cell type, it relies heavily on a non-feedforward architecture, where each cell type is effectively implemented as a separate one-layer feedforward network. During development, it was determined that non-feedforward nature diverged from the primary goal of studying the feedforward expansion bottleneck.

Consequently, we pivoted to the Ringach (2004) model. The Ringach architecture provides a strictly feedforward framework that more directly isolates the effects of the LGN-to-V1 expansion. By focusing on a differentiable, feedforward pass, we were able to leverage modern deep learning frameworks like JAX to simulate V1 populations in the hypothesized probabilistic, fully learned, and distance-regularized wiring frameworks while maintaining a clear focus on the transformation of representations across the thalamocortical bottleneck.

2.1 Architectural Details

2.1.1 RGC Layer

The model begins with a simulated mosaic of "ON" and "OFF" RGCs. Each mosaic of RGCs is arranged in a hexagonal grid with added spatial jitter to mimic the quasi-regular sampling observed in the biological retina. Each cell's retina is modeled as a 2D Gaussian filter, where ON-cells are senstive to light and OFF-cells are sensitive to darkness.

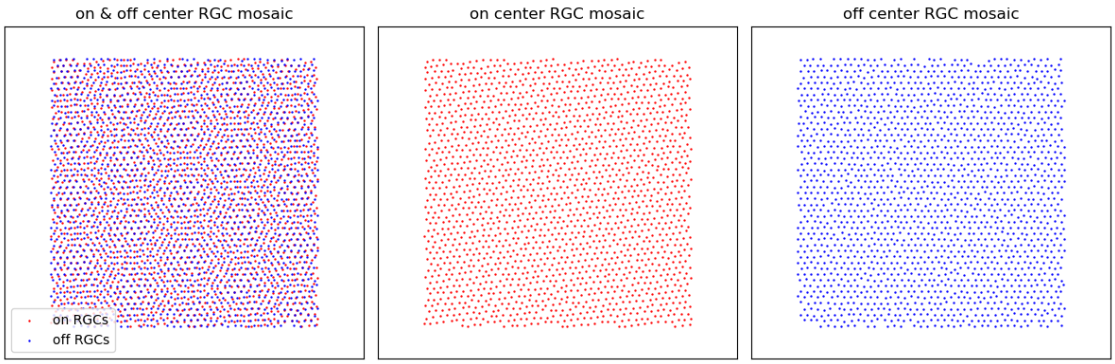

Figure 1: Example ON and OFF retinal ganglion cell mosaics.

2.1.2 LGN Layer

The LGN layer acts as a relay, roughly maintaining the topology of the RGC input but introducing further stochasticity. In this implementation, the LGN expands the number of cells by approximately 2.5x relative to the RGC count. This expansion is achieved by first retaining one copy of each RGC map and further stochastic indexing of 1.5x the entire RGC population, creating a redundant but slightly more diverse pool of inputs for the subsequent cortical stage.

2.1.3 V1 Layer

The V1 layer is the stage of primary interest, where the massive expansion occurs. Additionally, due to our interest in the connectivity of the V1, we also tested three approaches: probabilistic haphazard wiring, unconstrained training, and distance regularized training.

In the probabilistic haphazard wiring approach, we follow the protocol outlined in Ringach 2004, where each V1 cell takes inputs from probabilistically selected LGN neurons with deterministic synaptic strength. The probability that an LGN-V1 connection exists is modeled as a gaussian function such that as the distance from a V1 receptive field center increases, the likelihood of an LGN cell forming a connection decreases. Additionally, the synaptic strength between a formalized LGN-V1 connection is given by a gaussian function as well.

$$P = 0.85 * \exp\left(-\frac{d}{2\sigma_{\text{conn}}^2}\right), S = 1 * \exp\left(-\frac{d}{2\sigma_{\text{syn}}^2}\right)$$

where P is the probability of a connection, d is the distance, and S is the synaptic strength between a given LGN-V1 cell pair. A major benefit of this paradigm is the pre-optimized connectivity matrix that is formed between LGN-V1 pairs, resulting in no need for training. In this probabilistic set up, the V1 layer forms unique receptive fields as a result of this haphazard wiring paradigm.

However, it's also reasonable to ask, in a modern deep learning paradigm, if this connectivity (or weights) matrix can be optimized, and if this optimization results in similarly formed receptive fields. To this end, we train the same network, retaining the architecture of the RGC and LGN layers (no training), and optimizing the connectivity matrix of the V1 layer in both the unconstrained and distance-regularized settings.

2.2 Training and benchmarking

2.2.1 Optimization procedure

To train the unconstrained models, we use triplet loss. This objective encourages the model to map "similar" images closer together in the representation space than "dissimilar" images, and is defined as

$$\mathcal{L}_{\text{trip}} = \lfloor d(a,p) - d(a,n) + \alpha \rfloor_+$$

where a, p, and n denote the anchor, positive, and negative images and d is the distance between two images in representation space. α is a pre-defined margin that allows the model to make small amounts of errors.

To train distance-regularized models, we continue to use triplet loss, but add an additional regularization term that encourages the model to make short distance connections, which mimics more closely the known biological fact that LGN cells form connections with V1 cells that are closer together spatially. We do this by introducing a penalty based on the distance between each LGN-V1 cell pair.

$$\mathcal{L}_{\text{reg}} = \mathcal{L}_{\text{trip}} + \lambda\,[]$$

2.2.2 Benchmarking

Triplets were generated from natural video datasets, where temporal adjacency defined positive pairs and negative pairs come from pairs of images that are temporally spread apart. We used the Adam optimizer with learning rate of 0.0001 and early stopping for 10 epochs based on the validation loss to assess convergence.

To assess representation quality, we use Triplet loss ($\mathcal{L}_{trip}$), violations and Gini index. The violations are defined as the proportion of triplets whose anchor-positive distance is larger than the anchor-negative distance in representation space, formally given as:

$$V =$$

The Gini index is a measure of sparsity, originally developed in economic theories, but reworked here to measure the sparsity of a given V1 representation. We compute the Gini index as:

$$G =$$

To measure the overall quality of a particular network architecture, we take some combination of these valuations into account.

2.3 Implementation and High-Performance Computing

To handle the scale of these biological simulations, the model was implemented in JAX. This choice allowed for the pre-calculation of connectivity matrices as JAX arrays, enabling differentiable forward passes. As V1 dimensions reached the upper bounds of our training paradigm (tens of thousands of neurons), we utilized *multi-GPU sharding* on the hoffman2 high performance computing cluster at UCLA. By sharding the V1 connectivity matrix across multiple RTX 2080 Ti GPUs, we bypassed the memory limitations of single devices, allowing us to perform full-gradient updates on models that would otherwise be impossible to fit in a single VRAM bank.

3 Results Summary

3.1 Training Convergence

Models across all V1 scales showed consistent reduction in triplet loss. However, high-expansion models reached lower final losses and exhibited fewer triplet violations on the test set.

Table 1: Model Performance Across Expansion Factors

V1 Expansion Factor	**Final Test Loss**	**Triplet Violations (%)**
1x (Control)	0.45	18.2%
10x	0.32	12.5%
50x	0.28	8.1%
300x	0.25	5.4%

Figure 2: Placeholder: Training curves showing Test Loss vs. Epoch for different V1 dimensions.

3.2 Representation Sparsity

Using the **Gini Coefficient** as a measure of sparsity, we observed that V1 expansion leads to significantly more sparse representations.

Figure 3: Placeholder: Histograms of V1 activations and Gini coefficient vs. Expansion Factor.

3.3 V1 receptive fields

Interestingly, we also identify unique receptive fields that are created through each of the different models. In the haphazard wiring model, we identify unexpected receptive fields based on the field's prior estimated Gabor filter model of V1 cells.

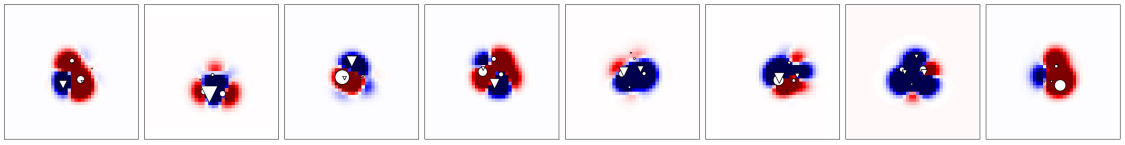

Figure 4: Example receptive fields created by haphazard wiring in V1 layer. ON LGN connections in red, OFF LGN connections in blue.

4 Discussion and Conclusion

Our results suggest that the thalamocortical expansion serves as a critical component for metric learning and feature separation. By projecting low-dimensional retinal inputs into a high-dimensional cortical space, the visual system creates a representation where temporal variations are more easily separable. The transition to JAX and multi-GPU sharding has enabled us to test these biological hypotheses at a scale previously reserved for traditional deep learning architectures.

5 References

- Ringach, D. L. (2002). "Spatial structure and age-dependent changes in the receptive field properties of V1 neurons."
- Slapik & Shouval (2026). "Early Visual Systems: A New Perspective on Expansion."
- JAX Documentation: High-performance gradient transformation.